

Adaptive Token Pruning: Boosting Cross-Lingual Transfer in Low-Resources

1. Problem Statement

Large multilingual language models (LLMs) provide strong performance across many languages but typically require billions of parameters and high computational resources. This makes them difficult to deploy in regions where low-resource languages are spoken and hardware infrastructure is limited. Many communities that speak languages such as Urdu, Swahili, and Quechua cannot benefit from modern NLP systems because the models are too large to run on edge devices like smartphones. This research investigates efficient cross-lingual transfer methods that allow compressed multilingual models to maintain strong performance while being small enough for local deployment.

2. Motivation

The digital divide in language technologies prevents many linguistic communities from accessing modern AI tools. Most multilingual models are optimized for server-scale environments rather than mobile or edge devices. By developing efficient compression and transfer methods, it is possible to deploy high-performing NLP systems locally without relying on cloud infrastructure. This work aims to enable “mobile-first” language models that support low-resource languages while maintaining reasonable accuracy and efficiency.

3. Objectives

The main objectives of this research are:

- Compress a multilingual language model from approximately 7B parameters to about 1B parameters.
- Maintain at least 85% of the original model’s performance on a selected low-resource language.
- Enable efficient inference on mobile or edge devices.
- Improve cross-lingual knowledge transfer using a teacher–student distillation framework.
- Evaluate performance on translation and summarization tasks.

4. Technical Contributions

This research proposes several technical innovations:

- **Adaptive Token Pruning:** Removing low-impact tokens during training to reduce computational complexity.
- **Dynamic Vocabulary Expansion:** Introducing language-specific tokens into a pretrained model without full retraining.
- **Pivot–Shadow Knowledge Distillation:** A framework where a large multilingual model transfers reasoning abilities to a smaller student model.
- **Low-bit Quantization:** Applying 4-bit quantization to reduce memory requirements and enable efficient inference.

5. Dataset(s)

The following datasets will be used:

- **FLORES-200**: A multilingual evaluation dataset covering more than 200 languages.
- **Common Voice**: A multilingual speech and text dataset for diverse language coverage.
- **Local News Corpus**: Web-scraped news articles in the target low-resource language for domain adaptation.

6. Baseline Papers / Methods

Baseline comparisons will include:

- **Mega-BAP: Benchmarking and Advancing Polyglot LLMs (2024)**
<https://arxiv.org/pdf/2311.07463v2>

7. Proposed Methodology / Flow

The proposed approach consists of the following pipeline:

1. Select a pretrained multilingual model as the teacher model.
2. Expand the tokenizer with language-specific vocabulary for the target language.
3. Apply adaptive token pruning to reduce unnecessary tokens and model complexity.
4. Train a smaller student model using cross-lingual knowledge distillation.
5. Apply 4-bit quantization to compress the model for mobile deployment.
6. Evaluate the compressed model on translation and summarization tasks.

8. Expected Outcomes

The expected outcomes of this research include:

- A compressed multilingual model approximately $7\times$ smaller than the original model.
- Retention of at least 85% performance on the target low-resource language.
- A mobile-first NLP model capable of running on edge devices.
- Improved accessibility of language technologies for underrepresented linguistic communities.